Dockerfile: FROM aur COPY

Tomcat, Nginx, Apache ka simple yaad rakhne wala notes

1. Teeno servers ka FROM aur COPY destination

Server	FROM	COPY ka destination
Tomcat	tomcat:10	/usr/local/tomcat/webapps/ROOT/index.html
Nginx	nginx	/usr/share/nginx/html/index.html
Apache	httpd	/usr/local/apache2/htdocs/index.html
WildFly	quay.io/wildfly/wildfly	/opt/jboss/wildfly/standalone/deployments/ROOT.war

2. COPY ka matlab

```
COPY  index.html  /usr/share/nginx/html/index.html
       ^                  ^
   tumhari file     container me final path
```

Pehla path = tumhare project folder ki file (Dockerfile ke saath wali).
Doosra path = container ke andar jahan server file dhundhta hai.

3. Sirf last word yaad rakho

Server	Folder ka naam	Port (container)
Tomcat	webapps/ROOT	8080
Nginx	html	80
Apache	htdocs	80
WildFly	deployments	8080

4. Poore Dockerfile

Tomcat

```
FROM tomcat:10
COPY index.html /usr/local/tomcat/webapps/ROOT/index.html
```

Nginx

```
FROM nginx
COPY index.html /usr/share/nginx/html/index.html
```

Apache

```
FROM httpd
COPY index.html /usr/local/apache2/htdocs/index.html
```

WildFly (yaha index.html nahi, **WAR file** copy hoti hai)

```
FROM quay.io/wildfly/wildfly
COPY ROOT.war /opt/jboss/wildfly/standalone/deployments/ROOT.war
```

Dhyan rakho: Tomcat/WildFly Java servers hain. WildFly seedha index.html nahi chalata, pehle app ko **.war** me banana padta hai. **ROOT.war** naam rakhoge to app seedha `localhost:8080/` par khulegi, warna `/naam` lagana padega.

5. Run karne ke commands

```
docker build -t myapp .
docker run -d -p 8080:80 myapp     # Nginx / Apache
docker run -d -p 8080:8080 myapp   # Tomcat / WildFly
```

Browser me kholo: **http://localhost:8080**

6. Chhote tips

• Destination me sirf folder likho (end me **/**) to file ka naam wahi rehta hai:
`COPY index.html /usr/share/nginx/html/`
• Naam badalna ho: `COPY mypage.html /usr/share/nginx/html/index.html`
• Port format: `-p laptop_port : container_port`